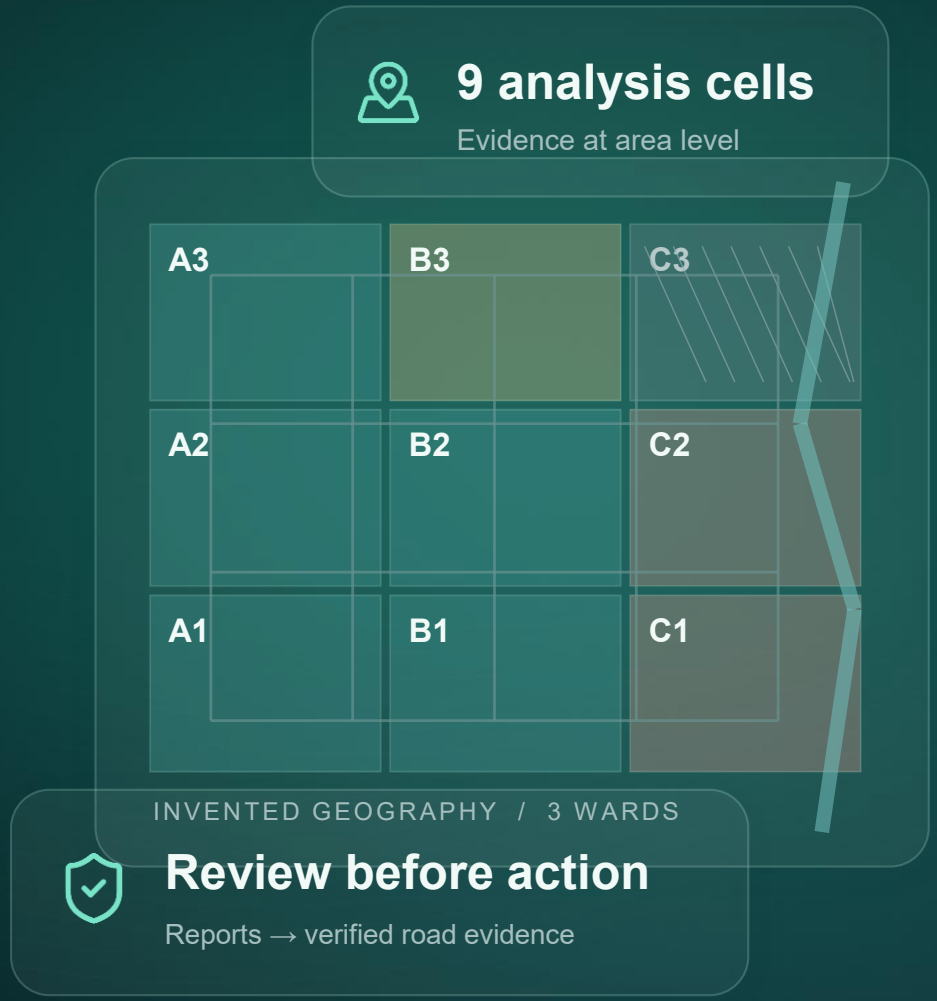
FLOODLINE

From rainfall to local decisions.

Smart City Flood Prediction
& Community Alert Platform

SYNTHETIC PILOT / v0.1.1

12 November 2026



Limited data has to become a useful local decision.

Residents and municipal reviewers need location, evidence and a supported next step.

01



Where is it vulnerable?

Rainfall, terrain and drainage need a shared geographic context.

02



What has been observed?

A report needs a time, a location and a reviewable account of conditions.

03



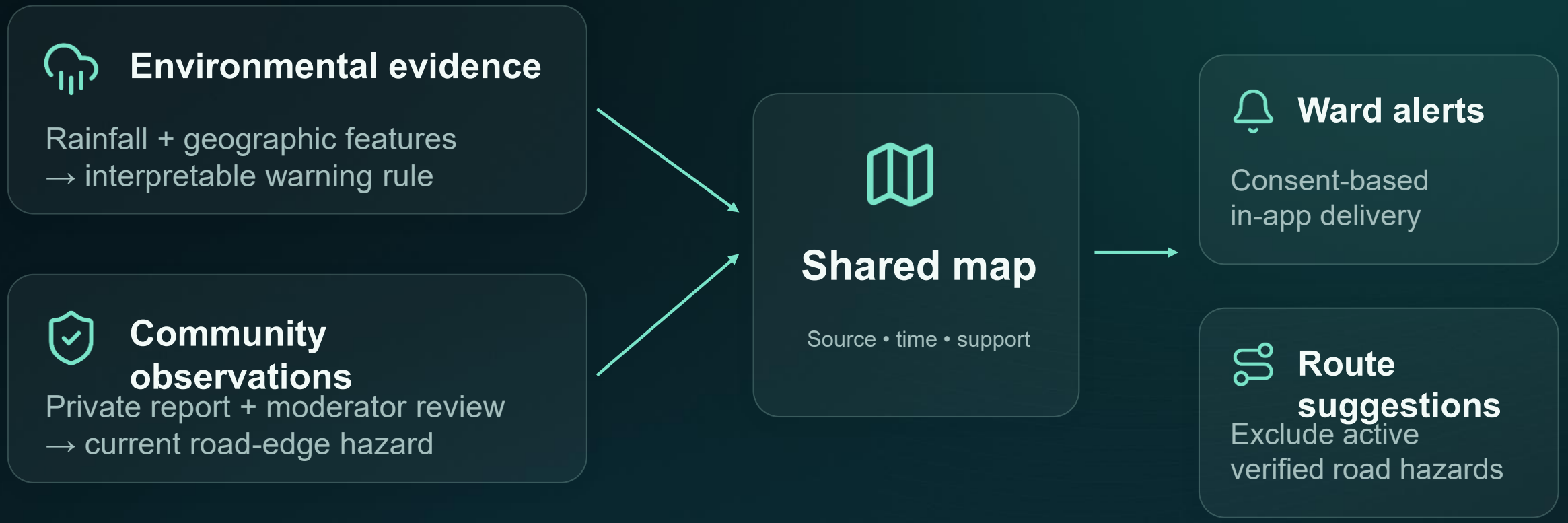
Which action is supported?

Warnings and routes must reflect current, verified evidence.

Floodline brings these decisions into one traceable workflow.

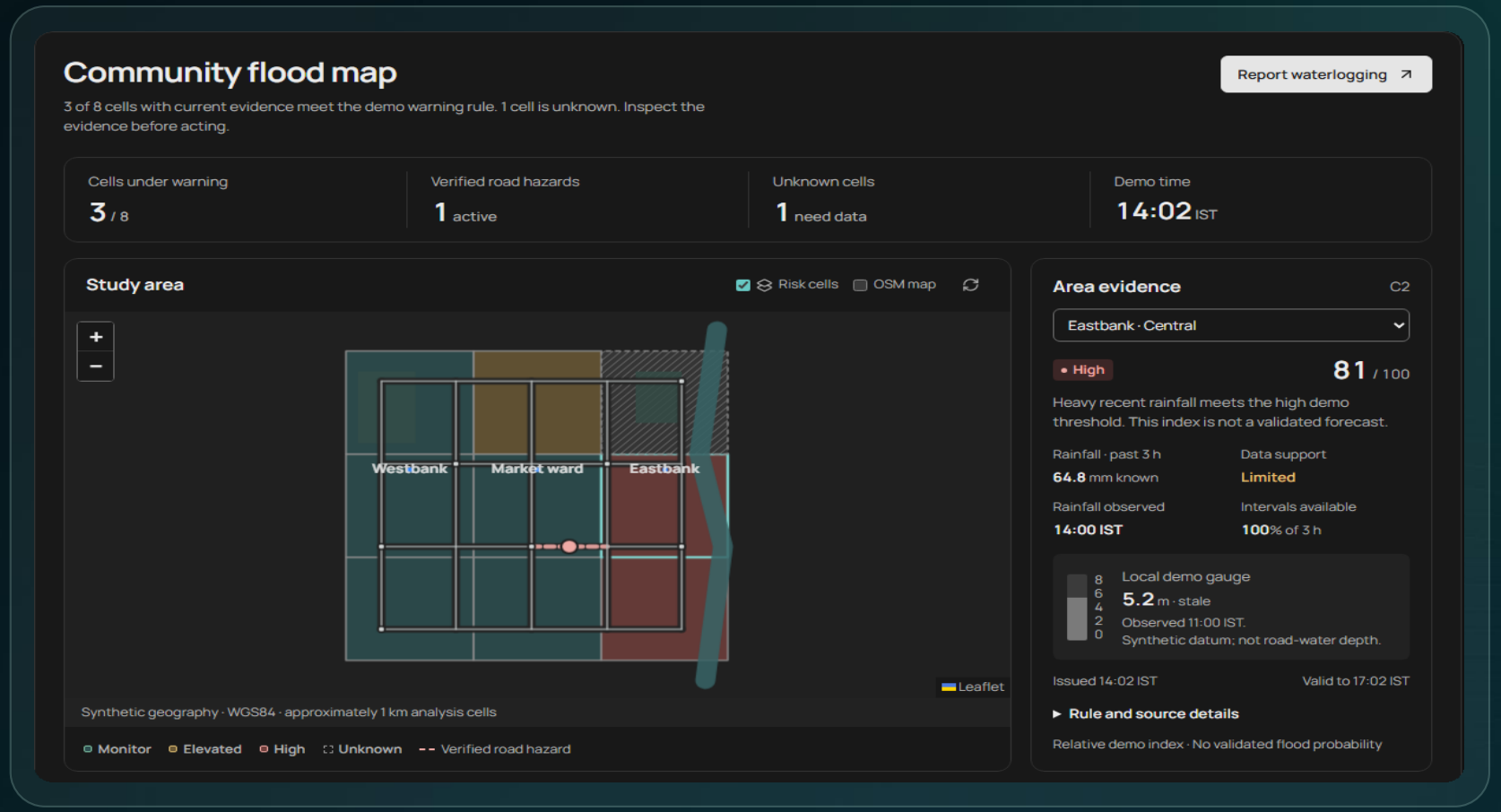
One platform. Two evidence streams.

Environmental warnings and observed road conditions keep their own meanings.



The map is the starting point.

A real capture of the delivered interface, with ward evidence beside the geography.



3
synthetic wards

9
analysis cells

61
directed road edges

Raw data becomes traceable evidence.

Checksums, explicit units and receipt-time availability make the input reproducible.

869 raw records

866

accepted observations

1 DUPLICATE

2 QUARANTINED

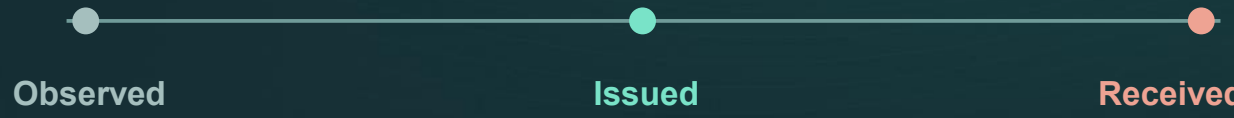
Validate

Units, ranges, timestamps and provenance

Preserve reasons

Invalid records enter a quarantine ledger

What was available at issue time?



This late record is excluded from the earlier snapshot.

GIS connects area context to a directed road graph.

The illustration uses the supplied synthetic geometry and road connections.



9 analysis cells

Approximately 1 km spatial support

21 nodes / 61 directed edges

One-way and disconnected cases included

WGS84 for display
UTM 45N for distances

Area susceptibility and road-water depth require different evidence.

ACTIVE BASELINE

An interpretable rule runs the demo.

The current index combines static susceptibility with recent rainfall.

INDEX = 0.35 S + 0.65 R

Rounded relative warning index

S: static susceptibility

Imperviousness, drainage, relative elevation and distance to water

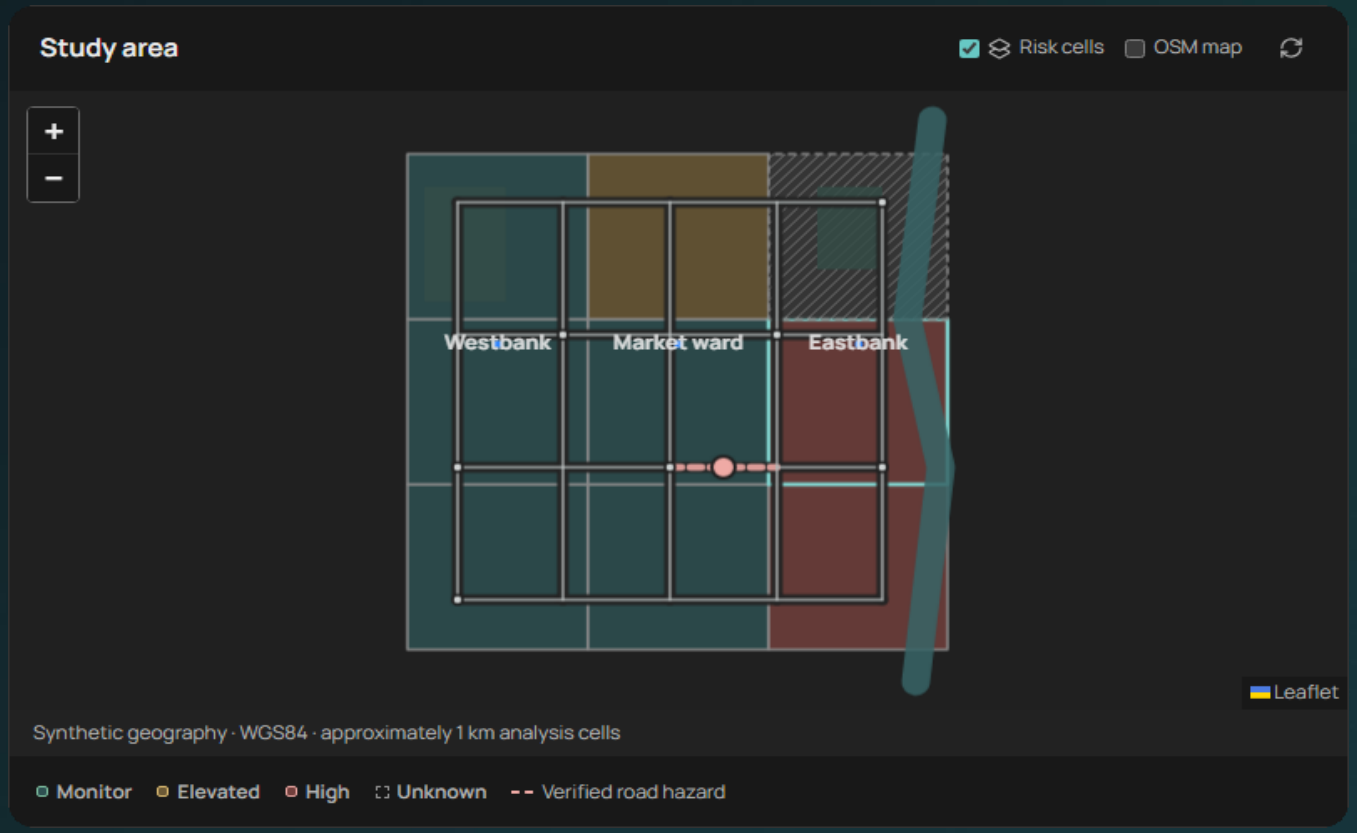
R: recent rainfall

Three-hour rainfall scaled to 75 mm, capped at 100

MONITOR	ELEVATED	HIGH	UNKNOWN
Index < 55	55–74	≥ 75	Missing / stale rain

The evidence is visible beside the map.

Risk words, timestamps and support flags help a resident interpret the display.



EASTBANK · CENTRAL

HIGH / DEMO RULE

81 / 100

64.8 mm rain / past 3 hours

Observed 14:00 IST • support: Limited

Relative index; no calibrated probability.

Reports earn trust through review.

An observation stays private while its time, location and publication permissions are checked.

RESIDENT / SUBMIT

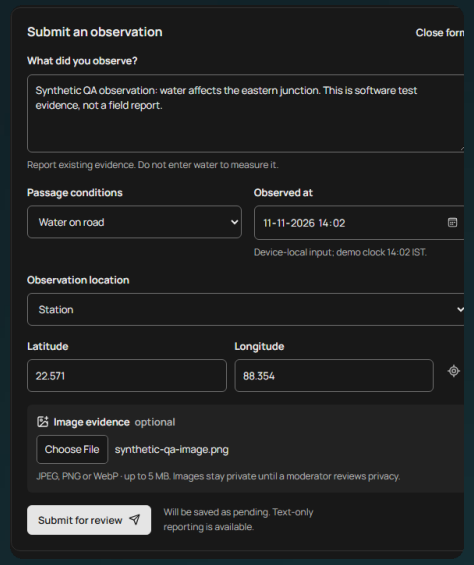
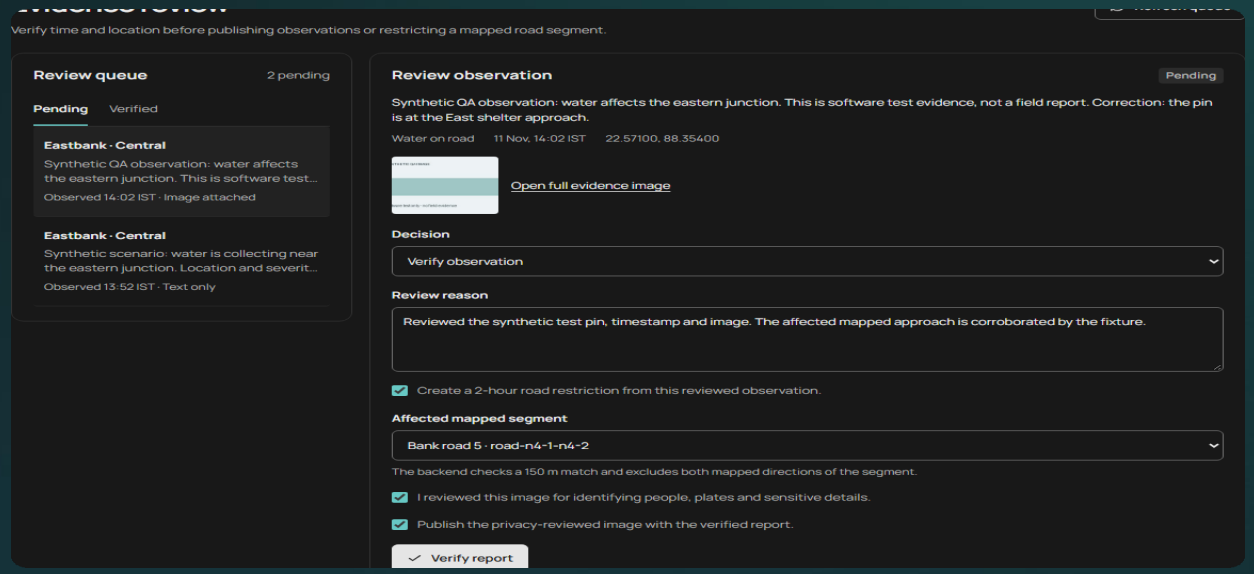


Image + location + observed time

Pending reports and images remain private.

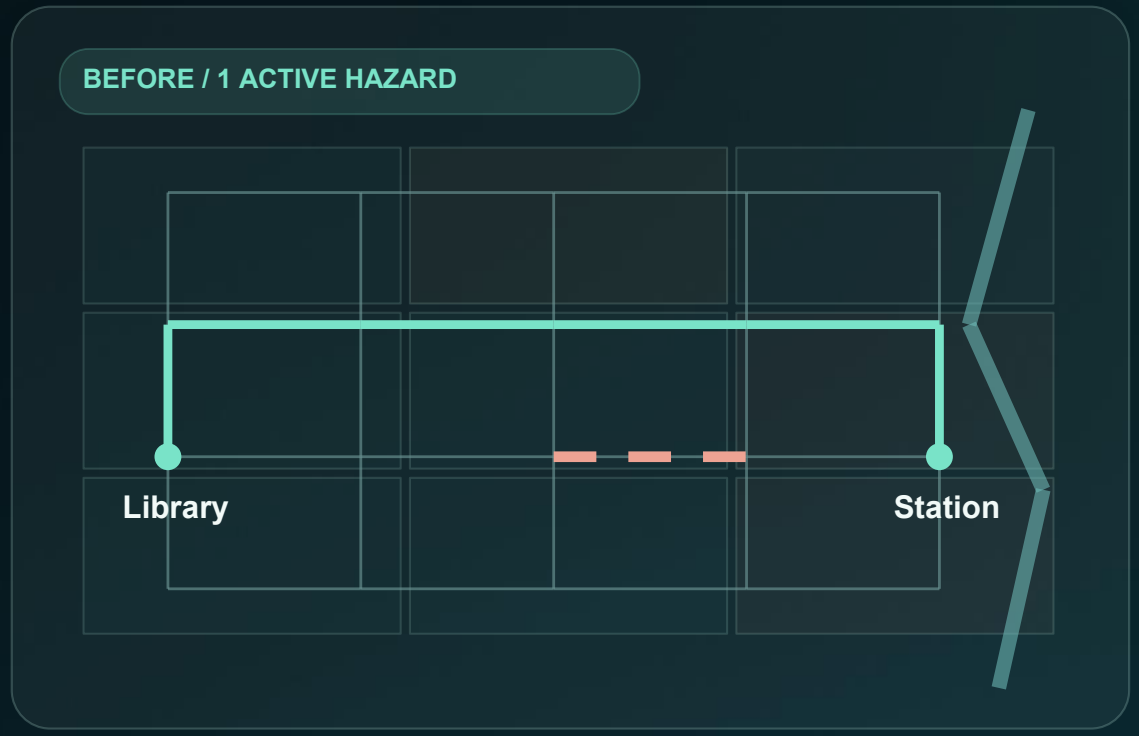
MODERATOR / VERIFY

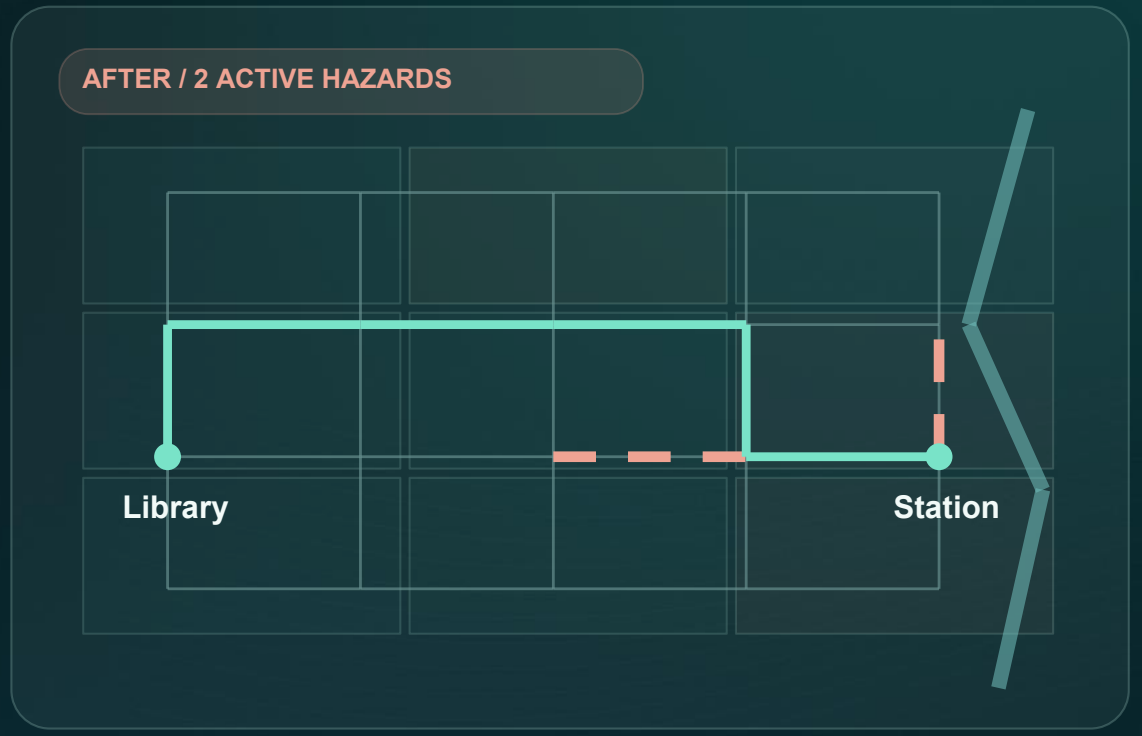


Authenticated review → bounded road hazard

Reviewed evidence changes the route.

The recorded Library → Station comparison excludes a newly verified segment.



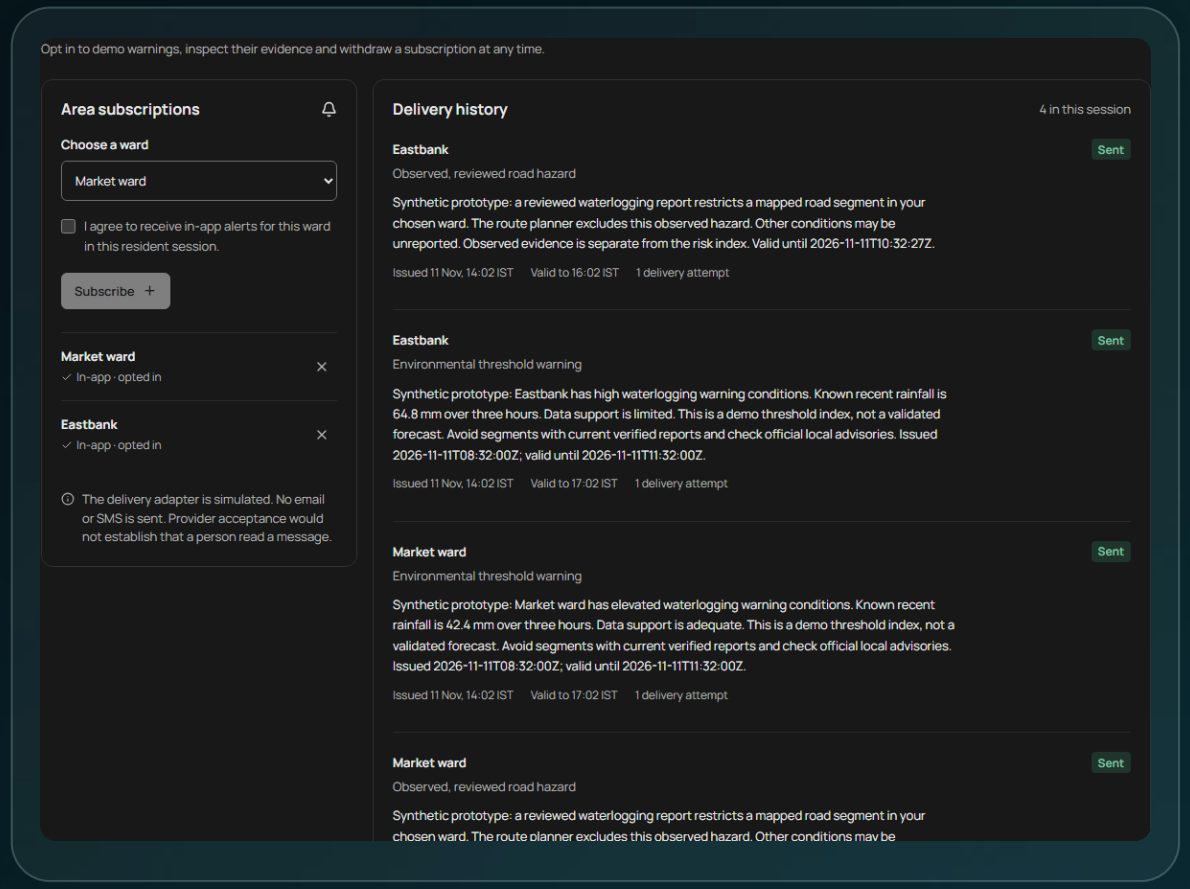


Different edges. Same 4.445 km.

The synthetic graph contains tied paths.
Only current, verified edge hazards are excluded.

Alerts remember consent, retries and expiry.

Persistent decisions prevent repeated processing from becoming repeated messages.



Opt in to a ward

A resident chooses subscriptions.

Deduplicate the trigger

Unchanged reminders use a 30-minute cooldown.

Bound delivery attempts

Retries after 30 and 60 seconds; 3 attempts maximum.

Cancel stale actions

Withdrawal, expiry and resolved hazards stop delivery.

Synthetic evaluation keeps events separate.

18 invented storms supply 624 usable rows; later storms remain held out.

TRAIN

10

events / 345 rows

Fit preprocessing

VALIDATE

4

events / 141 rows

Select F_2 threshold

TEST

4

events / 138 rows

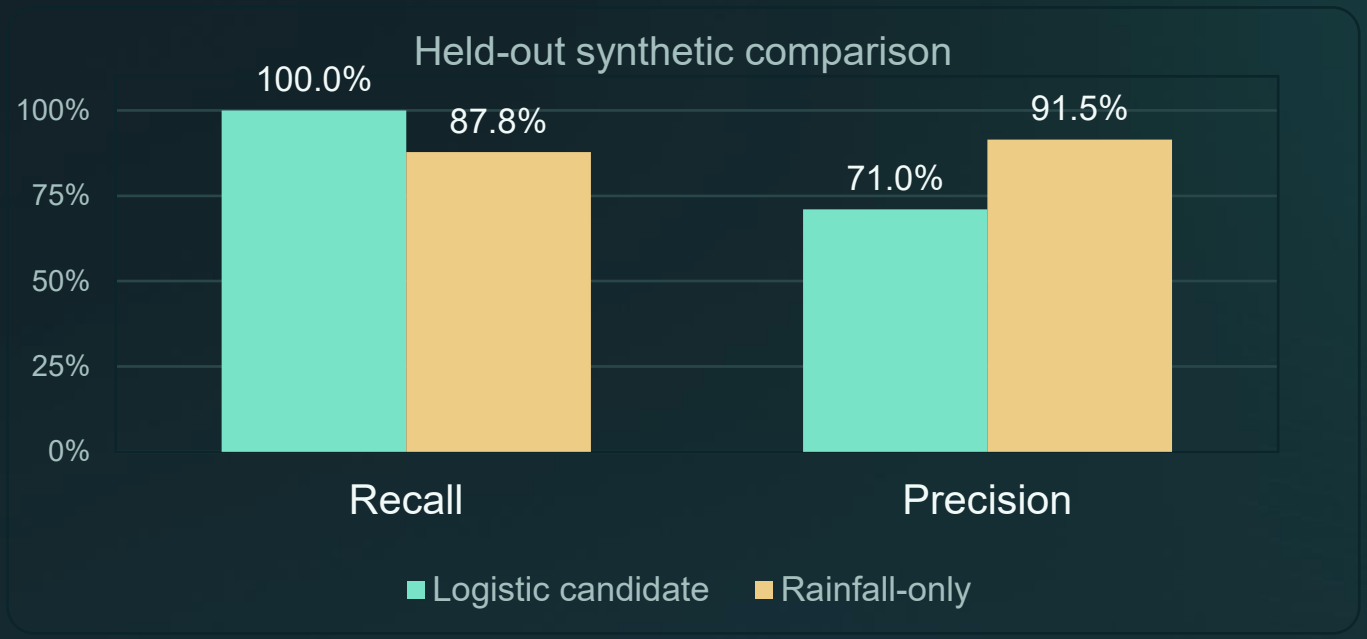
Freeze final comparison

Earlier events → later events. No row substitutes for an independent storm.

EVALUATION RESULT

Higher recall brought more false positives.

Held-out synthetic rows compare the logistic candidate with a rainfall-only baseline.



TEST-ROW ERRORS

	Logistic	Rain-only
False positives	40	8
Missed positives	0	12

138 rows / 4 test storms

Candidate remains unpromoted: it flags every test row.

Unknown data stays visibly unknown.

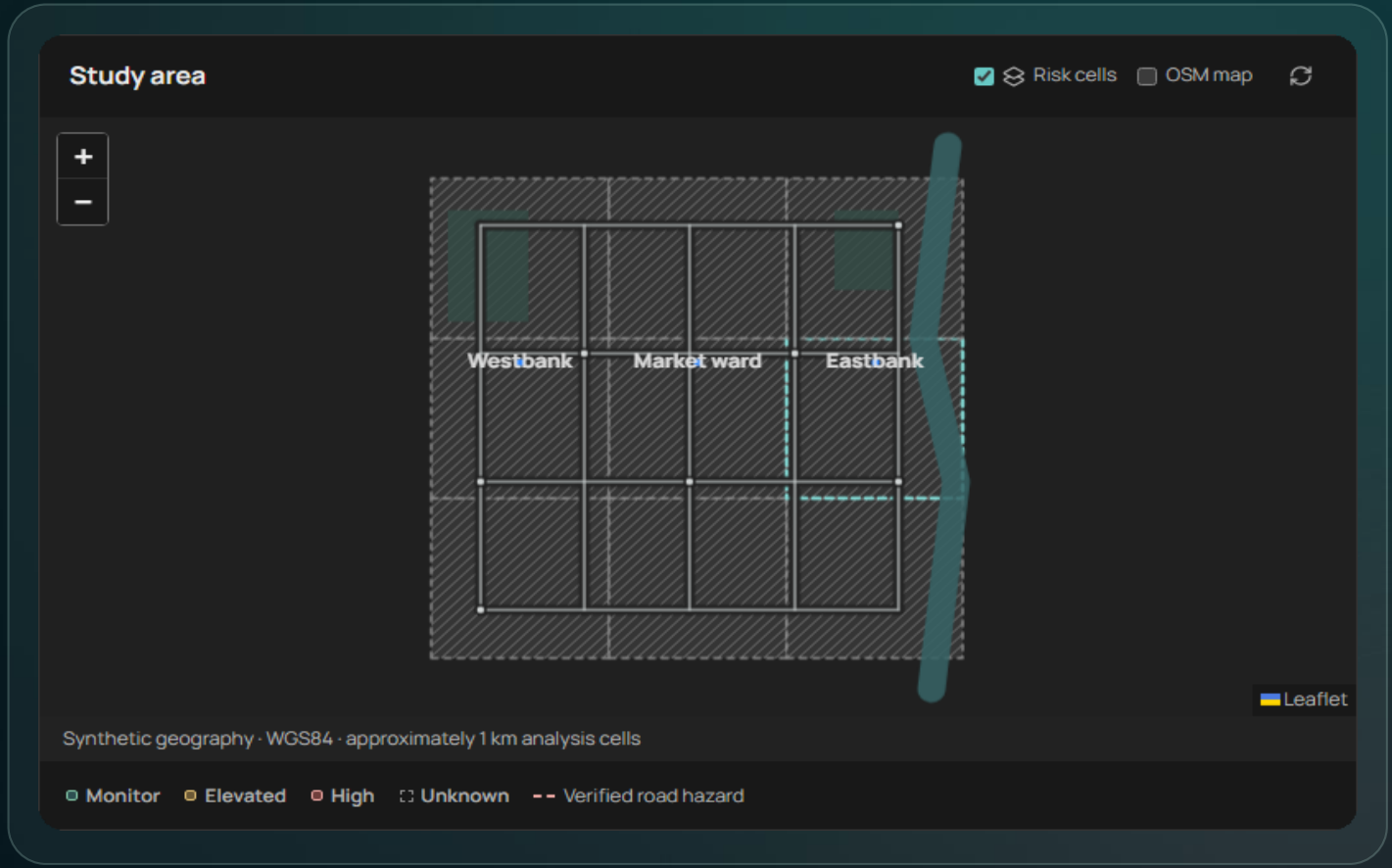
The outage scenario preserves original issue times instead of refreshing stale evidence.

9 / 9

cells become Unknown
during the recorded feed outage

Feed outage

No invented observations
and no zero-rain substitution



RECORDED ACCEPTANCE

The delivery is backed by inspectable checks.

Acceptance evidence for the synthetic v0.1.1 release, recorded 11 November 2026.

44
backend tests

14
browser workflows

7
Windows launcher checks

62
captured UI states

Light + Dark / phone to desktop

320–1440 px captures, keyboard flows, consent, moderation, routes, outage and recovery.

Local dashboard p95: 57.42 ms

10 clients / 20 loopback requests / SQLite. Fresh Linux CI also passes the automated workflow.

A focused demo follows six actions.

The same workflow can be shown live or explained using the preceding captures.

01 Inspect

Select an area and check its evidence.

02 Report

Submit time, location and an observation.

03 Verify

Review evidence and match a road segment.

04 Reroute

Compare changed edges after verification.

05 Subscribe

Opt in to a ward and inspect delivery history.

06 Replay

Run heavy rain, outage and synthetic recovery.

OPEN LOCAL DEMO

Launch start.bat beforehand. The console shows the selected address.

The path to a real pilot is clear.

The synthetic software workflow is delivered. Real-world readiness needs its own evidence.

Delivered now

- ✓ Local map and area evidence
- ✓ Private reports + moderation
- ✓ Verified road exclusions
- ✓ Consent-based alert lifecycle
- ✓ Evaluation and release records

SYNTHETIC DEMONSTRATION

Choose a pilot and licensed inputs

Audit source access, coverage and time semantics.

Validate with historical flood events

Use credible labels and separated held-out events.

Run field and human reviews

GIS, usability, local language and team sign-offs.

Accept the deployment environment

Test PostGIS, hosting and real notification delivery.

Visible evidence supports useful local decisions.

FLOODLINE

Trace the data. Review the observation. Explain the action.

- **Traceable inputs**
- **Reviewed reports**
- **Accountable actions**

Questions & discussion

github.com/Godavenger/floodline-caa

Six responsibility areas connect the stack.

M1–M6 are proposed team labels from the pipeline; named contribution records remain to be completed.

M1

Data and sources

Python • provenance • quality checks

M2

GIS and routing

Shapely • PyProj • NetworkX

M3

Baseline and evaluation

Interpretable rule • scikit-learn

M4

Backend and integration

FastAPI • SQLite • authenticated APIs

M5

Resident and moderator UI

React • TypeScript • Leaflet

M6

Alerts and acceptance

Delivery state • tests • release evidence

Requirements, implementation and evidence are linked.

Project-authored synthetic fixtures supply this release's data.

Project records

Assignment
project description.docx, item 3

Implementation plan
Smart_City_Flood_Prediction_CAA_Pipeline.md

Release and limits
PROJECT_REPORT.md / MODEL_CARD.md

Measured evidence
evaluation.json / acceptance records

Data provenance
SOURCE_REGISTER.md / data/manifest.json

Implementation references

FastAPI
Leaflet
scikit-learn
PostGIS

Real-pilot candidates

IMD • India-WRIS/CWC • NASA IMERG
Open-Meteo • municipal records • OSM

Access and suitability remain unverified.